



Published in final edited form as:

Drug Alcohol Depend. 2022 May 01; 234: 109394. doi:10.1016/j.drugalcdep.2022.109394.

Legislatively mandated implementation of medications for opioid use disorders in jails: A qualitative study of clinical, correctional, and jail administrator perspectives

Ekaterina Pivovarova^{a,*}, Elizabeth A. Evans^b, Thomas J. Stopka^c, Claudia Santelices^d, Warren J. Ferguson^a, Peter D. Friedmann^e

^aDepartment of Family Medicine and Community Health, University of Massachusetts Chan Medical School, 55 Lake Avenue North, Worcester, MA 01655, USA

^bDepartment of Health Promotion and Policy, School of Public Health and Health Sciences, University of Massachusetts Amherst, 312 Arnold House, 715 North Pleasant Street, Amherst, MA 01003, USA

^cDepartment of Public Health and Community Medicine, Tufts University School of Medicine, 136 Harrison Avenue, Boston, MA 02111, USA

^dClaudia Santelices, Institute for Health Equity and Social Justice Research, Northeastern University, 360 Huntington Ave, Boston MA, 02115, USA

^eUniversity of Massachusetts Chan Medical School (UMCMS) – Baystate and Baystate Health, 3601 Main Street, Springfield, MA 01107, USA

Abstract

Background: Individuals with legal involvement and opioid use disorders (OUD) are at an increased risk of overdose and premature death. Yet, few correctional systems provide all FDA approved medications for OUD (MOUD) to all qualifying incarcerated individuals. We report on the implementation of MOUD in seven Massachusetts' jails following a state legislative mandate to provide access to all FDA-approved MOUD and to connect with treatment upon release.

*Correspondence to: Department of Family Medicine and Community Health, University of Massachusetts Medical School, 55 North Lake Drive, Worcester, MA 01540, USA. ekaterina.pivovarova@umassmed.edu (E. Pivovarova).

Authors' contributions

Elizabeth Evans and Peter Friedmann are the multiple Principal Investigators, with all other co-authors serving as Co-Investigators. All co-authors Ekaterina Pivovarova drafted the manuscript. All co-authors provided comments and finalized and approved the submitted version.

CRedit authorship contribution statement

EE and PF conceptualized the study and obtained funding. All coauthors developed the protocols, collected data, supervised coding, and analyzed data. EP drafted the manuscript. All co-authors provided comments and finalized and approved the submitted version.

Author Disclosures

This manuscript presents original data that has not been published elsewhere and is not under consideration for publication in other journals. Each author has contributed significantly to all parts of this manuscript, reviewed the final manuscripts, and agrees with the submission to *Drug and Alcohol Dependence*. Parts of this manuscript were presented at the Addiction Health Services Research conference in November of 2021.

Conflicts of Interest

The authors have no conflicts of interest to declare.

Methods/participants: Based on the Exploration, Preparation, Implementation, and Sustainment framework, 61 clinical, corrections, and senior jail administrators participated in semi-structured interviews and focus groups between December 2019 and January 2020. Qualitative analyses focused on external and internal contexts and bridging factors.

Findings: Participants detailed how the outer context (i.e., legislative mandate) drove acceptance of MOUD and assisted with continuity of care. Salient inner context factors included decision-making around administration of agonist medications, staff perceptions and training, and changes to infrastructure and daily routines. Leadership was critical in flattening standard hierarchies and advocating for flexibility. System-based characteristics of incarcerated individuals, specifically those who were pre-sentenced, presented challenges with treatment initiation. Inter- and intra-agency bridging factors reduced duplication of effort and led to quick, innovative solutions.

Conclusions: Implementation of MOUD in jails requires collaboration with and reliance on external agencies. Preparation for implementation should involve systematic reviews of available resources and connections. Implementation requires flexibility from institutional systems that are inherently rigid. Accordingly, leaders and policymakers must recognize the cultural shift inherent in such programs and allow for resources and education to assure program success.

Keywords

Opioid use disorder; Medications for opioid use disorder; Medication assisted treatment; Jails; EPIS framework; implementation science

1. Introduction

Individuals with legal involvement and opioid use disorders (OUD) are at high risk for overdose and premature death (Binswanger et al., 2013, 2007; Pizzicato et al., 2018). Provision of medications for OUD (MOUD; methadone, buprenorphine, naltrexone) for this high-risk population decreases rates of mortality, improves intention to access and remain in community-based treatment, and decreases recidivism (Brinkley-Rubinstein et al., 2018; Curran et al., 2012; Evans et al., 2021, 2022; Malta et al., 2019; Paulozzi et al., 2015). Nonetheless, few programs provide MOUD to incarcerated individuals in the United States and limited research addresses how such programs are implemented and sustained (Bandara et al., 2021; Grella et al., 2020; Komalasari et al., 2021)

Currently, major national changes are underway, with significant resources earmarked for the implementation of MOUD in corrections (Weizman et al., 2021). Additionally, recent legal state precedents have established that prohibition of MOUD in jails and prisons violates the Eight Amendment and Americans with Disabilities Act (Sclafani v. Mici, 2020; Coppinger v. Pesce, 2018; Smith v. Aroostook County, 2021; Malta et al., 2019). Massachusetts, a state where the opioid-related overdose death rate is 120 times higher for formerly incarcerated persons than the general population, was the first in the nation to pass a legislative mandate to pilot provision of MOUD in jails (Massachusetts Department of Mental Health, 2017; Massachusetts General Laws, 2018). The law stipulates that MOUD treatment be maintained in individuals receiving it prior to detention and initiated

among sentenced individuals when appropriate. Jails must also facilitate continuation of medications in the community upon release.

Contextual factors influence implementation of health innovations and empirically based practices in legal settings, such as administration of MOUD (Friedmann et al., 2012, 2015; Mitchell et al., 2016; Taxman and Belenko, 2012). For example, transformative leadership, community partnerships, staffing, and funding are critical to MOUD delivery in jails (Ferguson et al., 2019). In a national study of 19 carceral systems that were early adopters of agonist medications (i.e., buprenorphine and methadone), federal regulations and program structures around agonist medication dispensation were important aspects of implementation (Bandara et al., 2021). These findings suggest that external (e.g., regulations around MOUD) and internal factors (e.g., program structure and leadership) impact acceptability and feasibility to administer MOUD in correctional agencies. While these studies begin to elucidate facilitators and barriers to program implementation and sustainment, they describe agencies that implemented MOUD early and voluntarily. However, most correctional agencies do not offer all forms of MOUD to all qualifying incarcerated individuals and staff in criminal legal settings have been documented as being resistant to MOUD, especially agonist medications (Andraka-Christou, 2021; Fiscella et al., 2018; Friedmann et al., 2015; Wakeman and Rich, 2018). Accordingly, understanding how programs respond to mandates for provision of MOUD and what factors impact implementation are necessary for translation to correctional agencies at large.

The current study examines qualitative findings from the first year of a legislative state mandate in jails to implement all FDA-approved forms of MOUD. Based on the Exploration, Preparation, Implementation, and Sustainment (EPIS) theoretical framework (Aarons et al., 2011), it offers theory-based empirical observations about the implementation of a comprehensive MOUD program.

2. Methods

2.1. Theoretical framework

The EPIS implementation science framework for public service programs was selected because it emphasizes inner and outer contextual influences and bridging factors on phases of implementation (Aarons et al., 2011). EPIS has been previously used to identify implementation factors associated with various substance use treatment modalities in legal settings (Aalsma et al., 2021; Becan et al., 2018; Garner et al., 2017; Zielinski et al., 2020). The outer context describes the impact of external policies, agencies, and environment on activities in the jails as they implement MOUD. The inner context includes factors inside the jails that affect acceptance and feasibility of MOUD administration. Bridging factors serve as inter and intra-collaborations and facilitate communication and cooperation between contexts, groups, and factors during implementation. These constructs, including the subheadings, were derived from the EPIS framework and adapted to this examination of the provision of MOUD in jails (Table 2 for definitions of individual constructs).

2.2. Participants and recruitment

Jail staff and on-site contractors at seven Massachusetts county jails, which expanded access to all FDA-approved MOUD in accordance with a legislative mandate, participated in semi-structured interviews and focus groups between December 2019 and January 2020. Participants were recruited by recommendation of jail administrators about key staff involved in decision-making around MOUD implementation. Staff received emails about the study and in-person meetings were held between staff and researchers describing the research. Participants were 61 clinical, corrections, and senior administrative staff (Table 1). Participation was entirely voluntary and all provided written informed consent prior to enrollment. The Baystate Health IRB served as the single IRB and approved all research procedures and materials.

2.3. Instrumentation

EPIS factors and prior research (Ferguson et al., 2019) informed development of interview guides for individual interviews, focus groups, and each type of key informants (see Appendix 1 for interview guide). Participants also completed a brief demographic survey.

2.4. Procedures

Interviews and focus groups were conducted by at least one investigator with qualitative research experience and a senior-level research coordinator. All interviews and focus groups were recorded, transcribed verbatim, and redacted for identifying information. Colloquialisms and utterances were removed to improve readability.

When feasible, senior staff were interviewed individually, with the other team members participating in focus groups. Focus groups ranged in size from 2 to 11 participants. All individual interviews were conducted in private spaces. Jail policies prohibited participant payments. Study participation was during regular work hours and participants were paid their normal wages.

2.5. Data coding and analysis

Coding used deductive (EPIS based) and inductive (ground up) approaches. Six investigators and six senior-level research coordinators used *a priori* codes from interview questions, the EPIS framework, and prior research to develop an initial codebook. Subsequently, transcripts were reviewed until new conceptual categories were exhausted. Preliminary codes were refined using constant comparative methods to define limits and boundaries of conceptual categories. Ten codes and thirty-five subcodes emerged. Six research coordinators, in three teams of two, double coded all transcripts using a segmentation approach (MacQueen et al., 1998). All transcripts were entered into Dedoose, a qualitative analysis platform (Dedoose., 2009). Each team of two reviewed all discrepancies between the coders until both coders agreed. If agreement could not be reached or questions remained, all coders and investigators reviewed the segment in question until there was complete agreement by the full coding team, including investigators.

The Framework Method (Gale et al., 2013) and EPIS guided data organization and analysis. Investigators and senior-level research coordinators reviewed codes until salient categories

emerged and thematic saturation was reached. Categories were grounded in EPIS constructs (see Table 2). Quotes were selected by the authors for accurate representation of themes. To enhance credibility of data analysis, the full research team, including jail collaborators, reviewed the results.

3. Results

3.1. Outer context

3.1.1. Service environment—Implementation of MOUD services in jails occurred amidst major national and statewide shifts towards greater acceptance of MOUD, leading to the passage of the legislative mandate. Participants uniformly described the mandate as driving organizational and individual acceptance, in large part because it originated at the highest levels of state government and not internal jail policy (see Table 2 for data examples). Contemporaneous with the mandate, a MA state court issued the first ruling of its kind that failure to provide methadone violated constitutional rights (Coppinger v. Pesce, 2018). Some participants interpreted this decision as a precursor to the inevitability of MOUD acceptance throughout corrections.

The most conservative sheriff...was dead set against this. 'Absolutely, positively not.' His director of Human Resources called me two weeks ago because they want to look into having a medication-assisted treatment program.[T]hey saw the writing on the wall.

3.1.2. Funding—The mandate allocated state funds to develop MOUD programming. Our participants observed that this appropriation helped not only with administration of MOUD in jails but also with continuity of care at release and with other agencies such as emergency rooms and inpatient substance use treatment facilities.

Most sites were in the early implementation stage and had not developed sustainment plans.

However, a site that had been offering agonist MOUD prior to the mandate described past difficulties in maintaining funding for continued provision, including having to repeatedly apply for grants to bridge funding gaps.

We have to negotiate with the legislature to get funding in order to do these kinds of programs. So, we also look for federal grants. We have [a federal grant] and an [MOUD] expansion grant that has helped us a little bit. But that's also a just a three-year grant so that money will end eventually too.

3.1.3. System-based characteristics of service recipients—A major external obstacle to providing MOUD in jails and within the guidelines of the mandate was managing individuals whose release dates were unknown or who could be transferred to facilities without MOUD access. The mandate required that all qualifying males be offered MOUD at least 30 days prior to release and that women be provided MOUD throughout their incarceration. However, our participants highlighted that for males, it was often impossible to determine when individuals would be released, and therefore when they should start medication.

[Starting MOUD] depends on...if they're sentenced, how long they have left in the program...[if] they can [get] bail...We have a very transient population there. For end of sentence gentlemen... the law says not less than 30 days to start them. Now we have parole dates and parole eligibility which could...half their sentence... [or] earned good time or work credits...So that 30-day mark isn't necessarily our 30-day mark.

Uncertain release dates, particularly for pre-trial individuals, impacted discharge plans and scheduling community follow-up as the mandate required. Another challenge with pre-trial individuals was not knowing where they would be incarcerated, if sentenced, and if that facility could continue their medication.

It's also hard...for pre-trial, we don't know where they're going and not every facility is doing [MOUD] right now. So, [starting] someone would actually be a disservice to them if they were...sentenced...at a facility that they wouldn't be continued on.

3.2. Inner context

3.2.1. Organizational characteristics—Participants explained that implementing MOUD in jails required a wide range of protocols around MOUD administration, as noted below:

[A] challenge from the start was how are we going to write policy and procedure for every piece of this. There are places in the facility where you don't need a policy. You need a policy when you're going to medicate a human being... whether it's the Narcan piece, the drug, the urinalysis piece, diversion response...we were gonna have to create or at least update existing policies for every corner of this program.

Perhaps the most complicated aspect of implementing MOUD was about how to and who would administer agonist medication (all sites had prior capacity to administer antagonist medication [i.e., naltrexone]). Specifically, methadone must be administered in a licensed facility that complies with a wide array of state and federal regulations (Substance Abuse and Mental Health Services Administration, 2015). Buprenorphine prescribing must be done by a physician who has received specialized training. Both methadone and buprenorphine have the potential for diversion, a major concern of correctional agencies (Evans et al., 2022). To comply with state and federal administration regulations around agonist MOUD, sites had to determine 1) if their in-house staff or contractors would provide methadone onsite or if they would transport individuals to a community-based opioid treatment program (OTP) and 2) if their staff or contractors would provide buprenorphine onsite. Jails engaged in extensive discussions about the benefits and challenges of each option, with few models available to guide them. They considered costs, facility infrastructure, staff knowledge and availability, state and federal requirements, security concerns, access to community providers, and relationships with contractors (see Table 3 on how sites provided MOUD).

Developing an onsite OTP required compliance with what were described as “onerous” state and federal regulations. Given these challenges, and length of time associated with acquiring

the OTP license, some sites chose to work with community-based providers. One interviewee observed that it was impractical to develop an in-house OTP for jails with multiple buildings as each building required a separate OTP, "...The regulations and the costs to have more than one dispensary are prohibitive."

However, relying on community-based providers for methadone also presented challenges. Jails had to either transport incarcerated individuals to and from the clinic daily, requiring multiple officers and nursing staff to be away from the facility, or develop procedures for community-based providers to bring and administer methadone onsite. Additionally, participants described difficulties medicating individuals on days when they were out of the facility (e.g., in court). For those that established OTPs in-house, transforming jail units into OTP facilities was complicated and slow. Participants described infrastructure problems for medication dissemination and the need for large spaces where dosing could be closely observed. Some ultimately converted previously unused rooms, whereas others used hallways and more secluded parts of the unit.

All facilities administered buprenorphine onsite, with some bringing in contractors. One issue that arose in working with contractors was delineating staff roles.

[W]e're working on trying to define roles because all of them overlap a lot...Ours is particularly interesting because we have four different vendors working together on this. So, it's been difficult...for everyone to...stay in their lane but also to help each other out.

Jail schedules are highly routinized and movement of residents is restricted. Thus, administering medications had to occur on a very tight schedule defined by security needs. Comprehensive procedures were developed to prevent medication diversion. Dosing was lengthy and, in some jails, occurred multiple times per day to accommodate the number of individuals requiring medications and their needs.

Interviewees expressed concerns that internal organizational factors such as restrictions on program capacity would ultimately limit their ability to provide MOUD to all individuals in need of treatment.

[M]y fear is that somehow, our capacity to treat more people is going to be capped...It's a lot to move these people off of a normal movement schedule. There's a lot of safety considerations that go into it...

3.2.2. Fidelity monitoring—Jails had to provide extensive MOUD implementation data to the state oversight agency. Participants described these reporting requirements as duplicative and taxing to the already overburdened staff.

The [data reporting] form that they [oversight agency] came out with. We're already mandated to do [a lot]...[and] getting a lot more put on us with staffing ... The questions that are asked on the [data reporting] forms are asked so many times in multiple forms. I just don't have the staff to do it. I don't have the capacity.

However, interviewees also recognized the importance of these data, especially if it ultimately helped to retain funding for this program.

3.2.3. Organizational leadership—Buy-in from senior leaders and staff about administration of MOUD in jails was critical. Organizational leaders had to shift traditional topdown work hierarchies to an approach that empowered their employees and incorporated staff feedback. To develop good multi-sectoral working relationships, leaders had to make it their priority.

I have been working very hard...on breaking down those silos...It may kill me to do it. Right from the beginning I said that there will be a treatment faction, a medical faction, a security faction, and a classification factor - all involved in the discussion. And at the end of the day, those people will sit at the table, and they will communicate. Because if we do not have all of those pieces together to do it, we are not going to make it work.

Clinical, administrative, and correctional leaders guided the transformation of highly structured correctional standards to allow for adaptation and flexibility necessary to incorporate staff perspectives.

[I]t's great to see how everybody's just being very flexible, we're coming in and we're very much disrupting a homeostasis. In a correctional facility everything runs by this time, this time, this time. And we come in, we're like 'no,' it just gets all thrown up. I really appreciate everybody mixing and starting to adapt to our tiny chaos.

3.2.4. Staff perspectives and hiring practices—Implementation of MOUD required a culture shift in how staff perceived agonist MOUD – from illicit substances to legitimate and necessary treatments.

A lot of the 30-year veterans, like myself...went from searching inmates for drugs and having it be the 'us and them,' 'cops and robbers' mentality to giving inmates [MOUD]. And they really had to wrap their heads around it.

Participants reported there were individual differences between staff in how they perceived MOUD. Those newer to corrections and younger, who were impacted by the opioid crisis, appeared more accepting of medication as treatment.

There was a tension... with the correctional officers some of whom I would call 'The Old Guard' who had been here for a long time who didn't see the ways that possibly substance use affected their own lives...A lot of new, young people who... grew up in a generation has high exposure to opiates...I found the younger officers were much more open to the concept that this was a brain disease...and...that [medication] treatment could help people.

During preparation and implementation, jails provided education and training to their staff.

Participants reported that correctional staff were also not the only ones in need of training.

...I was shocked actually because I thought that [correctional staff acceptance] was going to be the barrier...[T]he barrier wound up being, really, the medical staff and the people that don't really have experience with [MOUD] and...training them to understand the gravity of it, the importance of it, the legitimacy of it.

For some staff, it was reassuring that medication was offered alongside psychosocial services and that treatment would continue in the community.

I heard people saying they felt that we were offering this medication as a substitute for another, like it could turn into another addiction. That's something I consistently heard when people didn't get much education surrounding it. Until they realized we're not just dependent on giving the client a medication. There's other components to it.

Overall, participants agreed that training on MOUD was necessary and had to be sustained, either continuously or as part of orientation.

3.3. Bridging factors

Communication and collaboration between external and internal contexts, as well as between different roles within the jails, served as bridging factors and were critical to implementation.

3.3.1. Inter-organizational networks—Participants discussed the importance of involving jail staff in decision-making by external agencies around the rollout of MOUD. Collaborations between jails and the state agency that oversaw the mandate fostered trust and identified practical limitations of providing MOUD in correctional settings.

[L]awmakers make laws, but they don't work in the trenches where those laws are going to be implemented. So, our [senior leader] and a couple of his close associates...said we want to be part of that conversation. And that helped. Because it gave them [the state] a view into our world.

Despite involvement of senior level administrators in the development of the mandate, some participants observed that correctional perspectives were still not sufficiently considered. They noted that the legislature did not appreciate “the amount of work that they were putting on us because they didn't understand” day-to-day jail operations. Additionally, participants frequently mentioned how little time and direct guidance they received in implementing a program of this scope.

During program preparation and implementation, senior personnel from all jails and the oversight state agency met monthly to discuss progress and share resources. Participants described these meetings and resulting collaboration as invaluable to implementing their programs. One participating jail had been providing MOUD for several years. Participants repeatedly referenced how beneficial it was to observe that jail's operations and review their guidelines in formulating their own procedures.

3.3.2. Intra-organizational collaborations—Participants repeatedly mentioned that implementation required and resulted in collaboration between disciplines that had previously been absent. This highlighted the uniqueness of jail settings where clinical services must occur within parameters established by security staff. Participants remarked that MOUD implementation produced a more cohesive jail environment.

We work really hard to have good collaboration across all the disciplines. So, sometimes as clinical staff, we know we have ideas that might not be as readily accepted by other disciplines, or by security. They might say, ‘Well, let’s weigh the pros and cons’. So, we started immediately having conversations and opening up the lines of communication.

Ongoing communication between disciplines resulted in acceptance and produced innovative solutions, “They’ve really thought outside of the box to make this work.” Multi-disciplinary teamwork helped to resolve problems quickly.

[Y]ou kind of have an idea of how it’ll unfold, but then once you start rolling it out, you notice things that you thought might be an issue are not an issue, and things that you thought were going to be fine become an issue... We’ve been good, especially during the multidisciplinary meeting to just iron out. And it’s quick. There’s not a lot of red tape... If you have a suggestion and the group’s in agreement, you can implement it pretty quickly.

3.3.3. Relationships with community based agencies—Participants remarked about the benefits of established relationships with community-based organizations. Jails consulted with these agencies about overcoming barriers (e.g., protocol development, infrastructure needs). Preexisting relationships with community agencies also reduced the burden of developing new contracts and orienting staff to the facility. One participant remarked, “One of the reasons why [another jail] was able to build their [methadone] dispensary earlier than us was because they had a relationship with [provider] first as their medical [provider].” Our participants also noted the importance of having a network of providers that could continue MOUD in the community.

[Individuals] are set up with an appointment [on release] with a Suboxone clinic. We do our best to get the appointment the day of release. The reason we want to get the appointment... that day or the next day is so there’s no gaps in treatment... If there was a gap in treatment what we did here is null.

4. Discussion

This study expands on a growing body of research about implementation of MOUD in corrections by describing the critical impact of outer and inner contexts and organizational bridging factors to the implementation of all types of MOUD in jails. Our participants detailed how external factors such as the legislative mandate and availability of funding drove program acceptance and helped with continuity of care across agencies and upon release. System-based characteristics of individuals incarcerated – those with unknown release dates and who might ultimately end up in facilities without MOUD also presented substantial challenges about when and if medications should be started. These findings have important implications for policy and practice to treat OUD in correctional settings across the nation.

These findings on inner context factors align with previous research by Bandara et al. (2021) and Komalasari et al. (2021) about the complexities of dispensing agonist

medications in correctional settings. Participants described important differences between administering methadone and buprenorphine, with pros and cons of sending individuals to community OTPs and using in-house staff versus contractors. Unlike earlier research, which described some aspects of medication dispensing as “chaotic” (Komalasari et al., 2021), our participants described rigorous policies and highly structured procedures. Consistent with Ferguson et al. (2019), leadership shaped implementation and acceptance. Committed leaders purposefully worked to flatten work hierarchies and advocated for flexibility. Similar to findings by Grella et al. (2020), Richard et al. (2020) and Wakeman and Rich (2018) we also observed negative attitudes toward MOUD by staff. However, rather than being role-specific (e.g., in correctional officer versus clinical staff), perceptions about MOUD were specific to individuals, with those newer to corrections and with personal knowledge being more accepting of MOUD. Like with previous research (Bandara et al., 2021; Grella et al., 2020), our participants acknowledged the importance of education and training, recommending that MOUD education become part of onboarding or ongoing training.

Inter- and intra-agency bridging factors were essential to implementation success. Participants detailed how relationships with other jails reduced duplication of effort and helped jails learn from one another, especially from sites that had already been providing MOUD. Regularly scheduled meetings with the oversight agency and other pilot jails helped with comradery and establishing procedures. However, even though some jail personnel participated in development of mandate guidelines, some staff believed that perspectives from corrections personnel were insufficiently considered, resulting in difficulties translating to daily jail operations. Collaboration across disciplines in jail was critical to resolving problems, which often required innovative solutions.

These findings should be interpreted in the context of several limitations. Data collection was conducted in the first three months of program initiation. Accordingly, we learned primarily about factors involved in preparation and during early implementation. Follow up data collection will examine adaptations to initial implementation, strategies for sustainment, and implementation outcomes. Interviewed individuals were key staff and likely invested in program success. Accordingly, they may have provided socially desirable responses. Still, participants reported on difficulties encountered, suggesting at least some willingness to disclose challenges. We identified potential participants in collaboration with sites in the study. As such, individuals who were not perceived as key informants may have different perspectives. Our study was conducted in one state in the northeast, with high degree of resources and support for MOUD. Qualitative research in other settings may yield differing findings in other regions with limited resources and support for treatment. Because we conducted focus groups, we were not able to examine differences in perspectives between roles and tenure, but future research may consider if those differences exist and how they would impact implementation. Finally, data collection was completed in January 2020 and findings do not reflect procedural changes that resulted from COVID-19 restrictions.

As for clinical implications, lessons learned from MOUD implementation thus far could be used to refine and sustain MOUD programming. This might include, for example, considering how to provide MOUD to everyone for whom it is clinically indicated no matter the length of time spent in custody or their release dates. It could also include developing

collaboration across clinical systems (in-house jail services, community providers, and consultants) to assure that treatment is provided in a timely manner and maintained upon release to the community. Lastly, we identified some of the essential factors for implementation of MOUD services in jails, which can aid efforts to implement similar clinical programming in other correctional settings.

We offer an implementation science empirical basis for consideration of external, internal, and bridging factors and their impact on implementing MOUD in jails. In implementing MOUD in their own correctional settings, agencies may want to consider the following. First, external pressures, such as legislative mandates effectively initiate implementation, promote acceptance, and result in changes in correctional settings (Caudill et al., 2014; Thompson and Mays, 1988; Worsley and Memmer, 2017). Mandates alone, however, are insufficient and require internal environments that accept them. Policy makers should also incorporate feedback from jails and allow for opportunities to change guidelines to assure that barriers can be addressed and reduced quickly. Second, implementation of MOUD in corrections involves collaboration with and reliance on external agencies (Emerson et al., 2021; Taxman and Sachwald, 2010). Accordingly, preparation for implementation should involve a systematic review of available resources, connections and effective local, regional or national models of delivery. Opportunities to enhance collaboration and communication among agencies and systems should be encouraged. Lastly, implementation of MOUD requires flexibility from correctional systems that are highly structured and security-focused, in part because of legitimate concerns about safety of individuals under their care and the public. Accordingly, leaders and policymakers must recognize the cultural shift inherent in implementing such programs and allow for resources and education to assure program success.

Acknowledgements

MassJCOIN Research Group: Baystate Health Randall Hoskinson Jr. Lizbeth Del Toro-Mejías Elyse Bianchet Patrick Dowd Tufts University School of Medicine: Rebecca Rottapel Allison Manco Mary McPoland Essex County Sheriff's Department: Sheriff Kevin F. Coppinger Jason Faro Franklin County Sheriff's Office: Sheriff Christopher J. Donelan Edmond Hayes Hampden County Sheriff's Department: Sheriff Nicholas Cocchi Martha Lyman, Ed. DHampshire County Sheriff's Office: Sheriff Patrick J. Cahillane Melinda Cady Middlesex County Sheriff's Office: Sheriff Peter J. Koutoujian Kashif Siddiqi Daniel Lee Norfolk County Sheriff's Office: Sheriff Jerome P. McDermott Tara Flynn Erika Sica Suffolk County Sheriff's Department: Sheriff Steven W. Tompkins Rachelle Steinberg Marjorie Bernadeau-Alexandre

Funding

The National Institute on Drug Abuse (NIDA) 3UG3DA044830-02S1, 1UG1DA050067-01, K23DA049953-01.

References

- Aalsma MC, Aarons GA, Adams ZW, Alton MD, Boustani M, Dir AL, Embi PJ, Grannis S, Hulvershorn LA, Huntsinger D, 2021. Alliances to disseminate addiction prevention and treatment (ADAPT): a statewide learning health system to reduce substance use among justice-involved youth in rural communities. *J. Subst. Abuse Treat* 128, 108368. [PubMed: 33867210]
- Aarons GA, Hurlburt M, Horwitz SM, 2011. Advancing a conceptual model of evidence-based practice implementation in public service sectors. *Admin. Policy Mental Health Mental Health Serv. Res* 38 (1), 4–23.

- Andraka-Christou B, 2021. Addressing racial and ethnic disparities in the use of medications for opioid use disorder: study examines racial and ethnic disparities in the use of medications for opioid use disorder. *Health Aff* 40 (6), 920–927.
- Bandara S, Kennedy-Hendricks A, Merritt S, Barry CL, Saloner B, 2021. Methadone and buprenorphine treatment in United States jails and prisons: lessons from early adopters. *Addiction* 116 (12), 3473–3481. [PubMed: 33999458]
- Becan JE, Bartkowski JP, Knight DK, Wiley TRA, DiClemente R, Ducharme L, Welsh WN, Bowser D, McCollister K, Hiller M, Spaulding AC, Flynn PM, Swartzendruber A, Dickson MF, Fisher JH, Aarons GA, 2018. A model for rigorously applying the Exploration, Preparation, Implementation, Sustainment (EPIS) framework in the design and measurement of a large scale collaborative multi-site study. *Health Justice* 6 (1), 9. [PubMed: 29654518]
- Binswanger IA, Blatchford PJ, Mueller SR, Stern MF, 2013. Mortality after prison release: opioid overdose and other causes of death, risk factors, and time trends from 1999 to 2009. *Ann. Intern. Med* 159 (9), 592–600. [PubMed: 24189594]
- Binswanger IA, Stern MF, Deyo RA, Heagerty PJ, Cheadle A, Elmore JG, Koepsell TD, 2007. Release from prison—a high risk of death for former inmates. *N. Engl. J. Med* 356 (2), 157–165. [PubMed: 17215533]
- Brinkley-Rubinstein L, McKenzie M, Macmadu A, Larney S, Zaller N, Dauria E, Rich J, 2018. A randomized, open label trial of methadone continuation versus forced withdrawal in a combined US prison and jail: findings at 12 months post-release. *Drug Alcohol Depend* 184, 57–63. [PubMed: 29402680]
- Caudill JW, Trulson CR, Marquart JW, Patten R, Thomas MO, Anderson S, 2014. Correctional destabilization and jail violence: The consequences of prison depopulation legislation. *J. Crim. Just* 42 (6), 500–506.
- Coppinger v. Pesce (2018) 335 F. Supp. 3d. <<https://casetext.com/case/pesce-v-coppinger>>.
- Curran GM, Bauer M, Mittman B, Pyne JM, Stetler C, 2012. Effectiveness- implementation hybrid designs: combining elements of clinical effectiveness and implementation research to enhance public health impact. *Med. Care* 50 (3), 217. [PubMed: 22310560]
- Dedoose LS, 2009. In: Eli Lieber TWaJT, editor.
- Emerson A, Allison M, Saldana L, Kelly PJ, Ramaswamy M, 2021. Collaborating to offer HPV vaccinations in jails: results from a pre-implementation study in four states. *BMC Health Serv. Res* 21 (1), 1–7. [PubMed: 33388053]
- Evans EA, Pivovarova E, Stopka TJ, Santelices C, Ferguson WJ, Friedmann PD, 2022. Uncommon and preventable: Perceptions of diversion of medication for opioid use disorder in jail. *J. Subst. Abuse Treat*, 108746 10.1016/j.jsat.2022.108746. Epub ahead of print. [PubMed: 35249789]
- Evans EA, Stopka TJ, Pivovarova E, Murphy SM, Taxman FS, Ferguson WJ, Bernson D, Santelices C, McCollister KE, Hoskinson R Jr., 2021a. Massachusetts justice community opioid innovation network (MassJCOIN). *J. Subst. Abuse Treat* 128, 108275. [PubMed: 33483222]
- Ferguson WJ, Johnston J, Clarke JG, Koutoujian PJ, Maurer K, Gallagher C, White J, Nickl D, Taxman FS, 2019. Advancing the implementation and sustainment of medication assisted treatment for opioid use disorders in prisons and jails. *Health Justice* 7 (1), 19. [PubMed: 31832801]
- Fiscella K, Wakeman SE, Beletsky L, 2018. Implementing opioid agonist treatment in correctional facilities. *JAMA Intern. Med* 178 (9), 1153–1154. [PubMed: 30073242]
- Friedmann PD, Hoskinson R Jr., Gordon M, Schwartz R, Kinlock T, Knight K, Flynn PM, Welsh WN, Stein LA, Sacks S, 2012. Medication-assisted treatment in criminal justice agencies affiliated with the criminal justice-drug abuse treatment studies (CJ-DATS): availability, barriers, and intentions. *Subst. Abuse* 33 (1), 9–18.
- Friedmann PD, Wilson D, Knudsen HK, Ducharme LJ, Welsh WN, Frisman L, Knight K, Lin H-J, James A, Albizu-Garcia CE, 2015a. Effect of an organizational linkage intervention on staff perceptions of medication-assisted treatment and referral intentions in community corrections. *J. Subst. Abuse Treat* 50, 50–58. [PubMed: 25456091]

- Gale NK, Heath G, Cameron E, Rashid S, Redwood S, 2013. Using the framework method for the analysis of qualitative data in multi-disciplinary health research. *BMC Med. Res. Methodol* 13 (1), 1–8. [PubMed: 23297754]
- Garner BR, Zehner M, Roosa MR, Martino S, Gotham HJ, Ball EL, Stilen P, Speck K, Vandersloot D, Rieckmann TR, Chaple M, Martin EG, Kaiser D, Ford JH 2nd, 2017. Testing the implementation and sustainment facilitation (ISF) strategy as an effective adjunct to the Addiction Technology Transfer Center (ATTC) strategy: study protocol for a cluster randomized trial. *Addict. Sci. Clin. Pract* 12 (1), 32. [PubMed: 29149909]
- Grella CE, Ostile E, Scott CK, Dennis M, Carnavale J, 2020. A scoping review of barriers and facilitators to implementation of medications for treatment of opioid use disorder within the criminal justice system. *Int. J. Drug Policy* 81, 102768. [PubMed: 32446130]
- Komalasari R, Wilson S, Haw S, 2021. A systematic review of qualitative evidence on barriers to and facilitators of the implementation of opioid agonist treatment (OAT) programmes in prisons. *Int. J. Drug Policy* 87, 102978. [PubMed: 33129135]
- MacQueen KM, McLellan E, Kay K, Milstein B, 1998. Codebook development for team-based qualitative analysis. *Cam J* 10 (2), 31–36.
- Malta M, Varatharajan T, Russell C, Pang M, Bonato S, Fischer B, 2019. Opioid- related treatment, interventions, and outcomes among incarcerated persons: a systematic review. *PLoS Med* 16 (12), e1003002. [PubMed: 31891578]
- Massachusetts Department of Mental Health, 2017. An assessment of fatal and nonfatal opioid overdoses in Massachusetts (2011–2015) August 2017.
- Massachusetts General Laws, 2018. An Act for Prevention and Access to Appropriate Care and Treatment of Addiction, Chapter 208, Section 98 <<https://malegislature.gov/Laws/SessionLaws/Acts/2018/Chapter208>>.
- Mitchell SG, Willet J, Monico LB, James A, Rudes DS, Viglione J, Schwartz RP, Gordon MS, Friedmann PD, 2016. Community correctional agents' views of medication-assisted treatment: examining their influence on treatment referrals and community supervision practices. *Subst. Abuse* 37 (1), 127–133.
- Paulozzi LJ, Strickler GK, Kreiner PW, Koris CM, 2015. Controlled substance prescribing patterns—prescription behavior surveillance system, eight states, 2013. *Morb. Mortal. Wkly. Rep. Surveill. Summ* 64 (9), 1–14.
- Pizzicato LN, Drake R, Domer-Shank R, Johnson CC, Viner KM, 2018. Beyond the walls: risk factors for overdose mortality following release from the Philadelphia Department of Prisons. *Drug Alcohol Depend* 189, 108–115. [PubMed: 29908410]
- Richard EL, Schalkoff CA, Piscalko HM, Brook DL, Sibley AL, Lancaster KE, Miller WC, Go VF, 2020. You are not clean until you're not on anything: perceptions of medication-assisted treatment in rural Appalachia. *Int. J. Drug Policy* 85, 102704. [PubMed: 32173274]
- Sclafani v Mici, 2019. U.S. District Court for Massachusetts <<https://dockets.justia.com/docket/massachusetts/madce/1:2019cv12550/217305>>.
- Smith v. Aroostook County, 2019. 19–1340, 1st Circuit <<https://law.justia.com/cases/federal/appellate-courts/ca1/19-1340/19-1340-2019-04-30.html>>.
- Substance Abuse and Mental Health Services Administration, 2015. Federal Guidelines for Opioid Treatment Programs. HHS Publication No. (SMA) PEP15-FEDGUIDEOTP Substance Abuse and Mental Health Services Administration, Rockville, MD.
- Taxman FS, Sachwald J, 2010. Offender supervision Imprint Willan.
- Taxman FS, Belenko SR, 2012. Implementing Evidence-based Practices in Community Corrections and Addiction Treatment Springer, New York.
- Thompson JA, Mays GL, 1988. State-local relations and the American jail crisis: an assessment of state jail mandates. *Rev. Policy Res* 7 (3), 567–580.
- Wakeman SE, Rich JD, 2018. Barriers to medications for addiction treatment: How stigma kills. *Subst. Use Misuse* 53 (2), 330–333. [PubMed: 28961017]
- Weizman JP, Manoff J, Baney T, El- Sabawi, 2021. National Snapshot: Access to Medications for Opioid Use Disorder in U.S. Jails and Prisons O'Neill Institute for National and Global Health Law at Georgetown Law Center.

- Worsley MK, Memmer A, 2017. Transparency behind bars: a history of Kansas jail inspections, current practices, and possible reform. *J. Crim. Justice Law* 1 (2), 71–88.
- Zielinski MJ, Allison MK, Brinkley-Rubinstein L, Curran G, Zaller ND, Kirchner JAE, 2020. Making change happen in criminal justice settings: leveraging implementation science to improve mental health care. *Health Justice* 8 (1), 21. [PubMed: 32892276]

Author Manuscript

Author Manuscript

Author Manuscript

Author Manuscript

Table 1

Qualitative interview/focus group participant characteristics, Massachusetts Houses of Correction, 2018–2019 (n=61).

Female, n (%)	37 (60.7)
Age, mean (sd)	45 (11)
Race, n (%)	49 (80.3)
White	5 (8.2)
More than one race	4 (6.6)
Missing	2 (3.3)
Asian	1 (1.6)
Black or African American	
Hispanic/Latino Ethnicity, n (%)	3 (4.9)
Education, n (%)	2 (3.3)
High school diploma or equivalent	6 (9.8)
Some college, but no degree	6 (9.8)
Associate's degree	13 (21.3)
Bachelor's degree	28 (45.9)
Master's degree	5 (8.2)
Doctoral degree or equivalent	1 (1.6)
Other	
Job title, n (%)	22 (38.6)
Behavioral health/addiction treatment	17 (29.8)
Administrative	11 (19.3)
Medical	6 (10.5)
Correctional	1 (1.8)
Other	
Years working in current position, n (%)	15 (27.3)
<1 year	14 (25.5)
1–3 years	18 (32.7)
4–9 years	7 (12.7)
≥10 years	1 (1.8)
Unknown	
Years working for your current agency, n (%)	7 (12.7)
<1 year	5 (9.1)
1–3 years	20 (36.4)
4–9 years	23 (41.8)
≥10 years	0 (0.0)
Unknown	

¹ Age missing for two participants.

² Roles missing for four participants.

³ Years in current position and jail missing for 7 participants.

Table 2Exploration, planning, implementation, sustainability (epis) constructs, definition, and representative quotes^a.

Constructs	Definition	Representative Quote
OUTER CONTEXT		
Service Environment	State and federal sociopolitical factors that impact implementation and delivery of MOUD in the service environment (i.e., jails).	[H]aving the legislat[ion] in place certainly helped... it wasn't just treatment people in the facility saying, 'this is what we are going to be doing.' It's now the government involved in saying this is an option that we want to offer to everybody.
Funding	Fiscal support provided by external systems necessary for implementation to occur.	...[N]ow [the jails] have money to provide these services...It's very good for the community.[Y]ou can continue this throughout the state...We don't have to worry about if a person goes from one place to the other. They don't have to interrupt their treatment.
Systemic-Based Characteristics of Service Recipients	Characteristics of individuals who are incarcerated, which are grounded in systemic issues, that impact delivery of MOUD.	[P]retrial was a big challenge, because these individuals... could be getting released at court. So, it was really important that we have, in the community, places that they could go same day, latest next day, for their next dose. So, that was one of our challenges, because we had to put a system in place so that if somebody does get released at court, do they know what they need to do? Are there the resources in the community for them to go to those places, drop in, and get their doses?
INNER CONTEXT		
Organizational characteristics	Structures and standard operational processes in jails that impact implementation of treatment.	In here, it's been interesting to adjust to different time frames for things. The assessment process... can be a couple of hours that we're meeting with somebody. So it's being able to find that time in the day, where they're not eating, where it's not freeze count, where there's not rec, they don't have a visit, or they're not at court... There's... all of these different factors that aren't factors in the community. So being able to be flexible and really learn all of those different aspects has been interesting, challenging and just something to get used to.
Fidelity monitoring	Procedures undertaken to ensure adherence to active delivery of MOUD.	If all of that [data required to be collected] is long term public health data to support funding for the same- it's killing us upfront. It's killing folks at the beginning of this. The forms are really geared towards a community-based setting, too. They're not geared towards a correctional setting.
Organizational Leadership	Characteristics and behaviors of individuals involved in oversight and decision-making about MOUD implementation in jails.	They're [senior leadership] open to suggestions, which helps a lot. And they'll come to each one of us. [Corrections captain] will come to each one of us, and say 'What are your thoughts on that? This is what MAT [is] suggested, what do you think? Do you think that would work?' And it does, it helps. It helps us down here [in the clinic] run things smoothly.'
Staff Perspectives and Hiring Processes	Description of staff perspectives on MOUD and organizational hiring processes around training of staff involved in the delivery of MOUD and its implementation.	You can tell the ones that ask a lot of good questions like they want to understand is a withdrawal or is that a detox is that the [delirium tremens]? ... They ask about medical. They come to you and they'll be interested in maybe going to trainings on their own or reading about what's going on... [W]hen they show that extra layer of interest and they approach staff...and you say, 'Okay... this is what's going on, they're not feeling well...' and they seem receptive, then those are the ones that you're picking.
BRIDGING FACTORS		
Inter-Organizational Networks	Relationships between organizations through which knowledge of MOUD is shared and MOUD goals about implementation are developed and established.	[I]t's been a real support. It's been great to be able to [say to other jails], 'Hey how are you guys gonna do this?' ...or we're dealing with [state agency]... 'How did you do it?' and trying not to reinvent the wheel. If someone got a policy from someone else, let's share it, so we can all just update it.
Intra-Organizational Collaborations	Structures and processes that exist and aid in development of relations between policy and practice entities, formal and informal influence and directives, and work of intermediaries.	I've been here 19 years, and the last year and a half, we've worked with more departments than I ever have. It's just been seamless. Like, I can go down to Programs or Classification or Security, and everybody – we're all on the same page. We know what the process is. We know what the plan is.
Relationships with Community Providers	Description of collaboration and working relationships with community based MOUD agencies.	Contracting, I don't think we would have gotten close to meeting the [mandate] deadline without bringing in that expertise.

^aAdapted from Aarons et al. (2011).

Table 3
Massachusetts JCOIN jails descriptions and medication for opioid use disorder (MOUD) provision details.

Site ID	Size of Facility	Rural/Urban	Type of MOUD	Methadone Provider and Location (Jail vs. contractor inside vs. community OTP)	Buprenorphine Provider (contractor vs. jail)	Naltrexone (contractor vs. jail)
1	Large	Suburban/Metro	Methadone, Buprenorphine, injectable Naltrexone	Contractor (Inside)	Contractor	Jail
2	Small	Suburban/Rural	Methadone, Suboxone, Subutex, Sublocade, injectable and oral Naltrexone	Jail	Jail	Jail
3	Large	Urban	Methadone, Buprenorphine, injectable and oral Naltrexone	Contractor (Inside)	Contractor	Jail
4	Small	Suburban/Metro	Methadone, Buprenorphine, injectable and oral Naltrexone	Community OTP	Jail	Jail
5	Medium	Suburban/Metro	Methadone, Buprenorphine, injectable and oral Naltrexone	Contractor (Inside)	Contractor	Jail
6	Medium	Suburban/Metro	Methadone, Suboxone, injectable and oral Naltrexone	Contractor (Inside)	Contractor	Jail
7	Large	Urban Methadone, Suboxone, Subutex, injectable and oral Naltrexone		Community OTP	Contractor	Contractor